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PALEOMAGNETIC AND RADIOMETRIC EVIDENCE FOR THE AGE OF THE COMORES ISLANDS, WEST CENTRAL INDIAN OCEAN

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Paleomagnetic measurements on 56 lavas and six K–Ar age determinations indicate that the Comores Islands, west central Indian Ocean, are no older than late Tertiary. The mean virtual geomagnetic pole for each island coincides with the present spin axis of the earth at the 95% confidence level. Because of the young age, the paleomagnetic data cannot resolve any north-south movement that may have occurred during the time interval recorded by the lavas sampled. The K–Ar ages from three of the islands range from 0.01 ± 0.01 to 3.65 ± 0.10 my. The radiometric results indicate that the islands increase in age to the southeast, a sequence consistent with the geomorphology. A magmatic center migrating northwest at about 7 cm/yr is consistent with the between-island age relationships, assuming that the K–Ar dates are reasonable estimates of the ages of the islands.

1. Introduction

Recent interest in western Indian Ocean tectonics has resulted in numerous controversies over the direction and timing of relative movement between plates. The role of Madagascar in Indian Ocean tectonics has proven to be an integral part of all interpretations. There are essentially three incompatible hypotheses involving Madagascar. Briefly, one proposal is that Madagascar was once 20° north of its present position attached to the sialic crust of the Seychelles Bank [1], requiring Madagascar to have a Tanzania-Kenya fit with Africa [2]. In contrast, Dixey [3] suggests that the Mozambique Channel between Africa and Madagascar is "geosynclinal" and underlain by Precambrian granitic crust; Flower and Strong [4] concur, basing their contention on some sandstone inclusions in some lavas of the Comores Islands. In addition, Madagascar may have occupied the southerly Inhambane delta region of Mozambique as suggested by Wright and McCurry [5] and discussed by Strong [6].

Additional constraints must be placed on the tectonic interpretations before contrasting hypotheses can be resolved. For this reason, a paleomagnetic investigation of the Comores Islands, located in the Mozambique Channel midway between Madagascar and Africa (fig. 1), was undertaken to determine if large-scale north–south movements had taken place

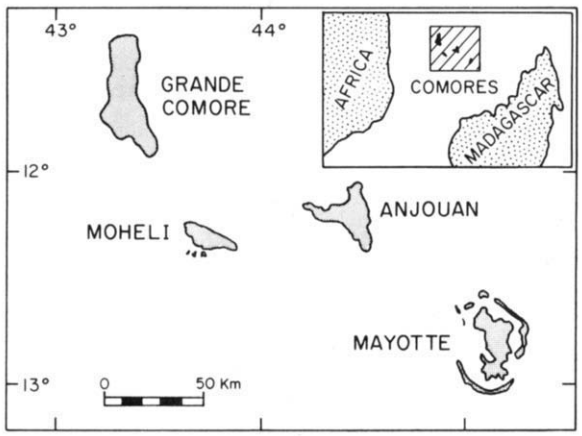


Fig. 1. Location map of the Comores Islands.

in the time interval recorded by the islands. K–Ar age determinations were made to verify the less precise age estimates indicated by the paleomagnetic data and to better delineate the between-island age relationships.

Esson et al. [7] have recently described the general geology and petrology of the islands. The geology of Grande Comore [8], Anjouan [9] and Moheli [10] has been described in detail. Earlier work [11, 12] is somewhat limited; however, De Saint Ours [12] suggested that volcanism on Mayotte took place in the Miocene and possibly as early as Cretaceous. Based on geomorphologic evidence, he [12] also proposed that the age of volcanism in the islands increased toward the southeast. Thus Grande Comore would be the youngest island and Mayotte would be the oldest.

2. Experimental procedures

Paleomagnetic cores of 2.5 cm diameter and 10 cm average length were obtained using a portable gasoline-powered drill [13]. The cores were geographically oriented while still attached to the outcrop. Approximately six cores were drilled along the same horizon and were separated from each other by 1–2 m. The samples were taken almost entirely from coastline exposures and roadcuts due to the lack of suitable exposures inland.

In the laboratory, a 2.3 cm length specimen was cut from the deepest, least-weathered part of each core for paleomagnetic analysis. The remnant magnetism was measured using a 5 Hz spinner magnetometer [14]; unstable magnetic components were minimized by application of peak alternating magnetic fields of 100, 200 and 300 Oe to each specimen. The mean magnetic direction was determined using the minimum scatter criterion of Irving, Stott and Ward [15]; this method is discussed by Watkins and Richardson [16]. It should be mentioned that the results do not differ significantly whether a single demagnetizing field treatment or a minimum scatter treatment is used. Standard methods [17] of data reduction and statistical analysis were used.

K–Ar dates were determined on whole rock chips from six lavas, one from Grande Comore, two from Anjouan and three from Mayotte. Armstrong [18] described the K–Ar analytical techniques in detail. K analyses were performed using a Model 303 Perkin–Elmer Atomic Absorption Spectrophotometer. Conventional isotope dilution procedures were used in the Ar analyses [18, 19].

3. Results

A total of 327 samples from 56 lava flows was collected for this study. Fig. 2 shows sample locations. The geologic units of De Saint Ours [12] are also shown on fig. 2.

Paleomagnetic data for each lava from Grande Comore, Anjouan and Mayotte are presented in terms of virtual geomagnetic poles (VGP's) in fig. 3. The mean VGP of each island and the 95% confidence limits are also shown in fig. 3. Table 1 summarizes the flow-average paleomagnetic data for each island.

The K–Ar results and the age-polarity relationships are given in table 2.

4. Discussion and summary

As illustrated in fig. 3, the mean virtual geomagnetic pole for each island coincides with the present spin axis of the earth at the 95% confidence level. If a fundamental paleomagnetic assumption that the geomagnetic field averages out to the spin axis over a minimum interval of about 10^5 yr [20] is correct, then the lavas sampled must be very young (Late Cenozoic) [21]. Because of the young age, the paleomagnetic data cannot resolve any north–south movement that may have occurred during the time interval recorded by the lavas sampled. Other implications of the paleomagnetic data are discussed elsewhere [22].

The K–Ar data (table 2) verify that the islands are indeed very young (Pliocene and younger) and that they increase in age toward the southeast, as suggested by De Saint Ours [12]. The sample locations (fig. 2) show that most of De Saint Ours' geologic units were dated. Since all of the ages obtained are younger than 4 my, it seems extremely unlikely that any of the exposed rocks are as old as Cretaceous, an age proposed by De Saint Ours [12]. The more extensive paleomagnetic data also support this conclusion.

It should be noted that if Madagascar has been in

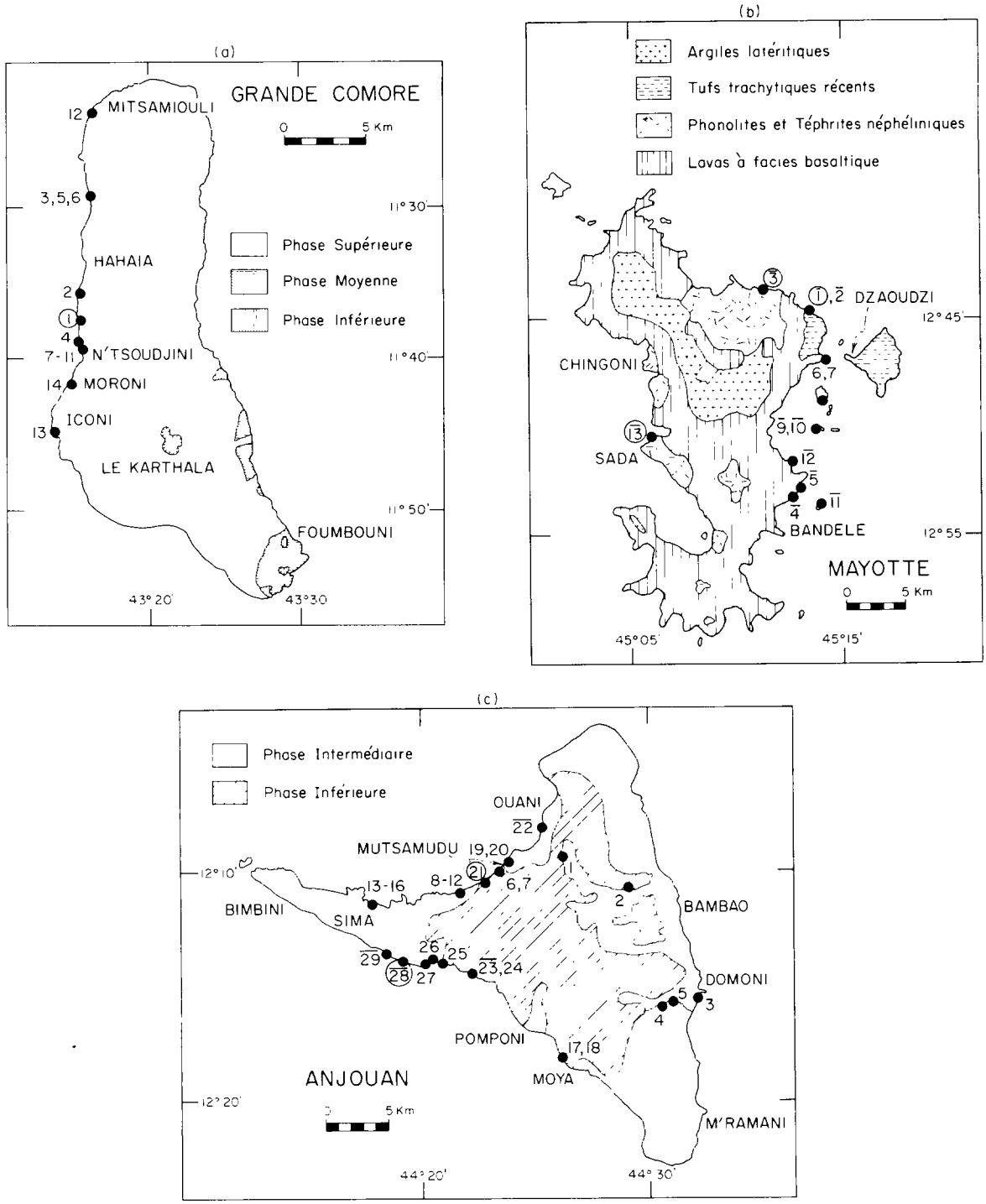


Fig. 2. Sample locations on (a) Grande Comore, (b) Mayotte and (c) Anjouan. Geologic units from De Saint Ours [12]. Bars above numbers indicate reversed magnetization, others are normally magnetized. Circled numbers represent dated samples.

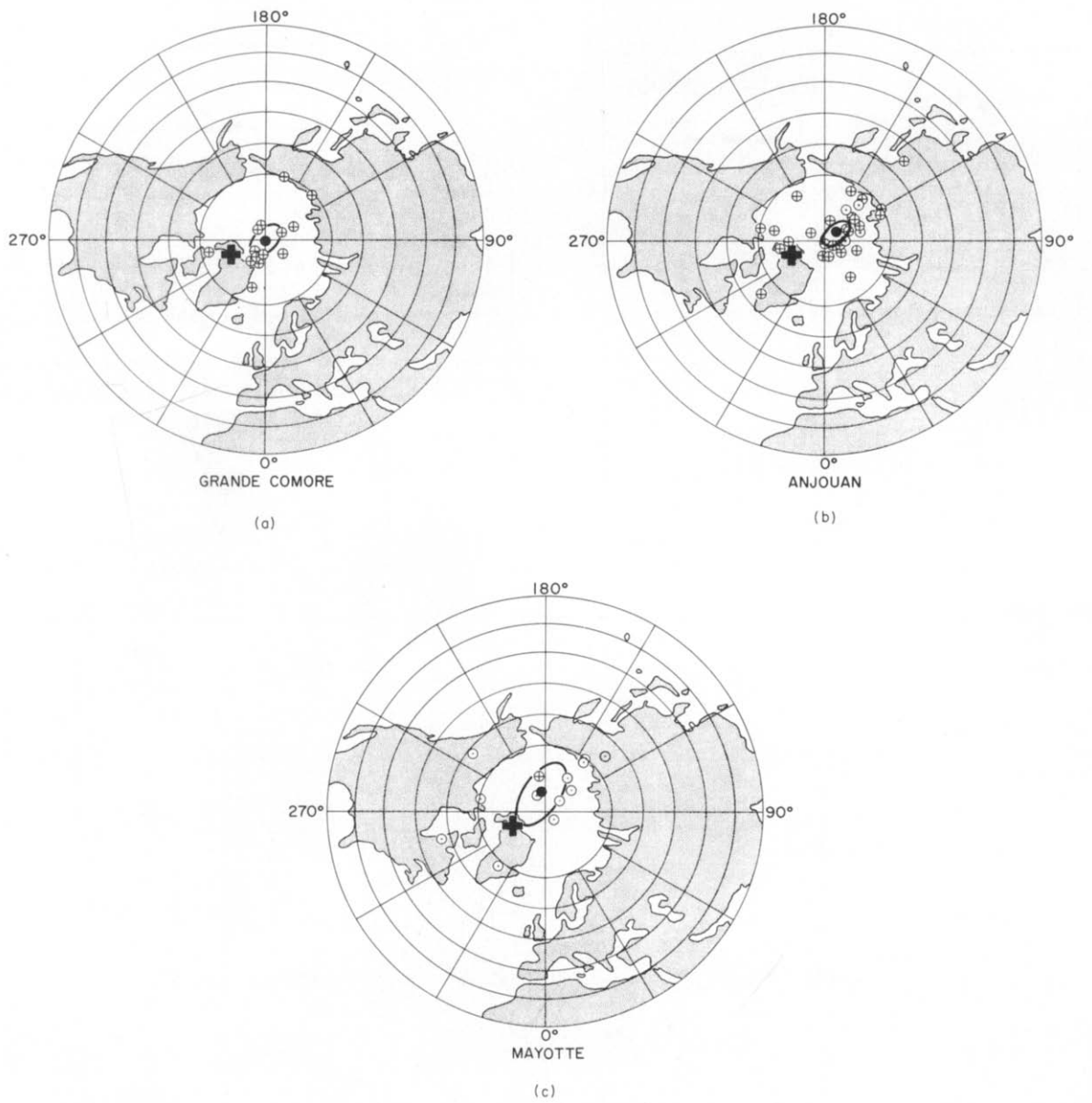


Fig. 3. Steriographic plots of VGP data from (a) Grande Comore, (b) Anjouan and (c) Mayotte. The large cross represents the position of the present magnetic pole. + = normal polarity, ○ = reversed polarity, ● = mean VGP of island. The oval represents the 95% confidence limits.

its present position relative to Africa since Cretaceous [24], then the young ages of the Comores Islands clearly preclude any close association with the inferred separation [1, 2]. By similar reasoning, it is not really surprising that a few sandstone inclusions [4] are found in some of the lavas. The inclusions merely show that there is some pre-Pliocene sediment beneath the islands.

The relative ages of the islands indicate that a magma source has migrated in a northwest direction along the trend of the archipelago at a rate of approximately 7 cm/yr, assuming that the K–Ar dates are reasonable estimates of the ages of the islands, and using 250 km, as the length of the archipelago. This is a rough estimate of the migration rate since the number of dates available gives only limited control

Table 1
Minimum dispersion flow-average data for each of the islands. Flow-average paleomagnetic data for (a) Grande Comore, (b) Anjouan and (c) Mayotte.

Island	N	D	I	R	K	α_{95}	θ_m	ϕ_m
Grande Comore	12	359.8	−21.6	13.6921	42.22	5.8	89.6	20.7
Anjouan	29	4.5	−22.0	28.0396	29.15	4.8	85.6	126.0
Mayotte	13	3.6	−32.6	12.3647	18.89	8.9	84.0	190.2

N = number of lavas sampled; D = magnetic declination in degrees east; I = magnetic inclination, (−) down, (+) up; R = length of the resultant magnetic vector; K = precision parameter equal to $(N-1)/(N-R)$; α_{95} = radius of the 95% confidence circle; θ_m = latitude of mean VGP, (−) south, (+) north; θ_m = longitude of mean VGP in degrees east.

Table 2
K–Ar results and polarity data; Decay constants: $^{40}K = 0.584 \times 10^{-10}$/yr, $^{40}K = 4.72 \times 10^{-10}$/yr; atomic abundance: $^{40}K/K = 1.19 \times 10^{-4}$.

Sample number	Magnetic polarity	Wt.% K	Radiogenic Ar STP (cm^3/g)	% Air Argon	Age (my) ± σ	Magnetic epoch
GC1 – 1	Normal	1.10	0.0031	100.9	0.01 ± 0.01	Brunhes
		1.11	0.0039	99.3		
AJ21 – 1	Normal	0.746	0.0134	98.2	0.39 ± 0.20	Brunhes
		0.737	0.0095	99.0		
AJ28 – 1	Reversed	0.681	0.0422	88.0	1.52 ± 0.10	Matuyama
		0.686	0.0404	89.0		
MY1 – 1	Reversed	1.62	0.199	98.3	3.4 ± 1.3	Gilbert
		1.62	0.234	98.0		
MY3 – 1	Reversed	0.810	0.0449	95.0	1.42 ± 0.15	Matuyama
		0.808	0.0468	92.0		
MY13 – 1	Reversed	3.02	0.432	89.0	3.65 ± 0.10	Gilbert
		3.04	0.454	75.0		

on the volcanic phases of the islands. An attempt was made to date the oldest stratigraphic unit exposed on each island. This was apparently achieved on Anjouan and Mayotte, but only the highest of three stratigraphic units [12] on Grande Comore was dated. Consequently, the calculated migration rate would be higher if the lowest stratigraphic unit on Grande Comore is significantly older than 0.01 my.

The Comores Islands are similar to the Hawaiian Islands and a number of other chains of volcanic islands in the Pacific Ocean in that they all show migration of volcanic activity along the chains. McDougall [23] has recently proposed that the observed age relationships in the Hawaiian Islands are due to movement of a magma source located in the asthenosphere. Since the movement of the magma source is opposite in direction to the plate motion, he suggests that the migrating magma source represents return flow of material toward the axis of spreading. Because the migration rate of the magma source is about twice the half-spreading rate of the associated oceanic rise, he concludes that counterflow must be taking place in a zone with a thickness comparable to that of the lithosphere. For the Comores Islands a spreading axis with a half-rate of about 2.5 – 3.5 cm/yr would be required to the north-west of the archipelago. It appears very unlikely that the African rift will satisfy this requirement, due to its very slow spreading rate [25], but one might propose that the counterflow is toward the Mid-Atlantic Ridge.

The precise mechanism operative during the formation of the Comores Islands is uncertain at the present state of knowledge. However, the very young age of the islands and the inter-island age relationships must

be consistent with any tectonic interpretation suggested for the region.

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